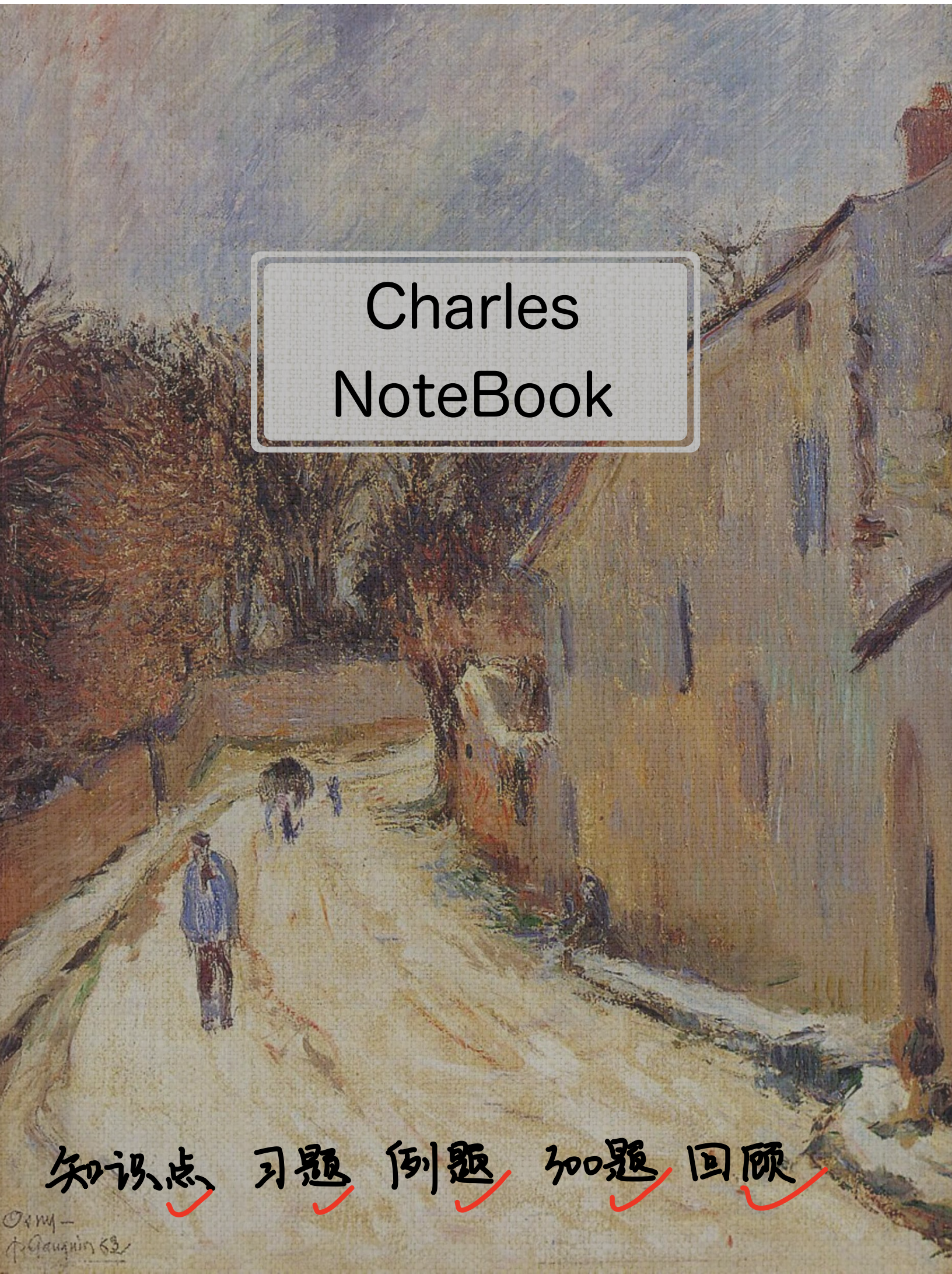
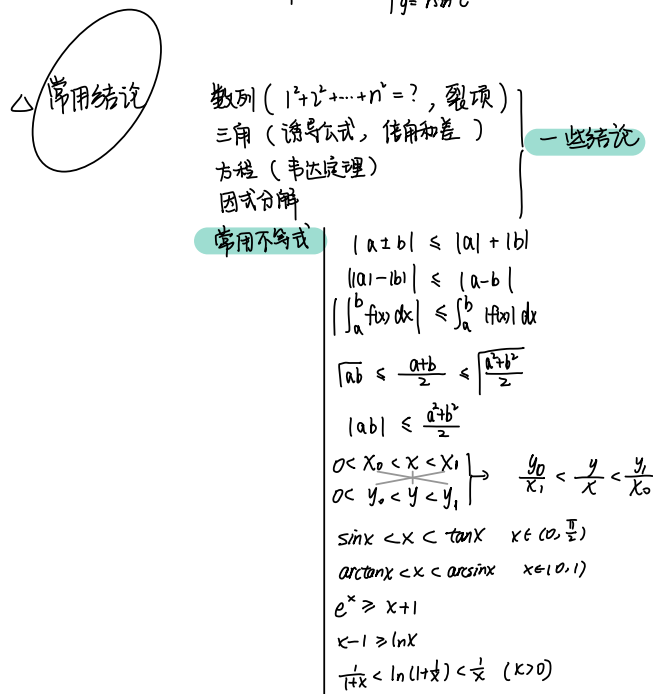
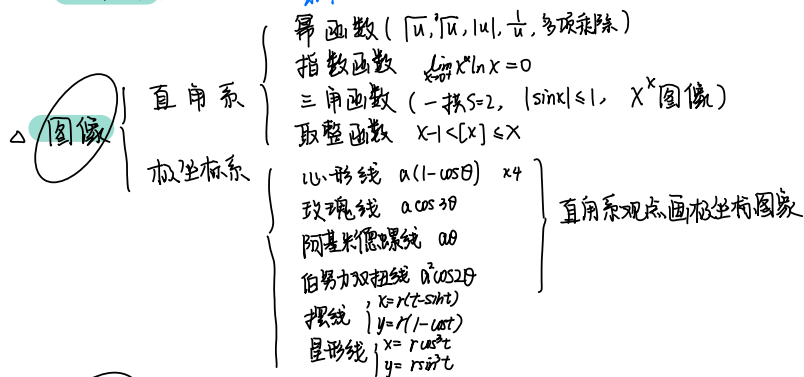
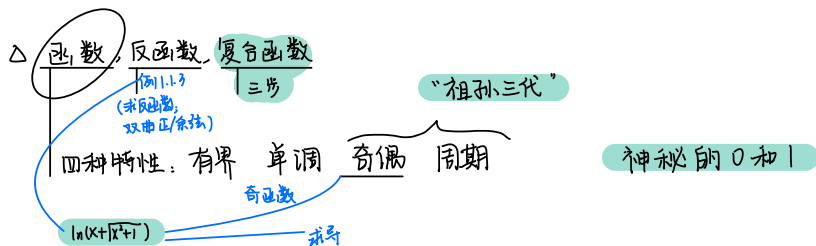




Charles NoteBook

知识点 ✓ 习题 ✓ 例题 ✓ 加题 ✓ 回顾 ✓



一、函数的概念与特性

例题

注意

① 函数，反函数 — $f[f^{-1}(x)] = x$ 逆反

反函数 $\begin{cases} \text{② } f'(x) > 0 / f'(x) < 0 \Rightarrow \text{有反函数} \Rightarrow \text{严格单调} \end{cases}$

$\begin{cases} \text{③ } y=f^{-1}(x) \text{ 关于 } y=x \text{ 对称} \\ \text{④ } y=f(x) \text{ 图像相同} \end{cases}$

例 1.1.3

求 $y=f(x)=\ln(x+\sqrt{x^2+1})$ 的反函数 $f^{-1}(x)$ 的表达式与值域.

$$\frac{x-\sqrt{x^2+1}}{x^2-(x^2+1)}$$

$$\begin{cases} y=\ln(x+\sqrt{x^2+1}) \\ -y=-\ln(x+\sqrt{x^2+1})=\ln\frac{1}{x+\sqrt{x^2+1}}=\ln(\sqrt{x^2+1}-x) \end{cases}$$

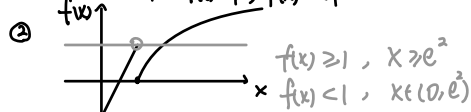
$$\begin{cases} e^y=x+\sqrt{x^2+1} \\ e^{-y}=-x+\sqrt{x^2+1} \end{cases} \quad x=\frac{e^y-e^{-y}}{2}$$

② 复合函数

- ① 把 $f(x)$ 的 x 换成 $g(x)$
- ② $g(x)$ 画图
- ③ 列表，写答案.

例 1.1.4 $f(x)=\begin{cases} \ln|x|, & x \geq 1 \\ 2x-1, & x < 1 \end{cases}$ 求 $f[f(x)]$

$$\text{① } f[f(x)]=\begin{cases} \ln|f(x)|, & f(x) \geq 1 \\ 2f(x)-1, & f(x) < 1 \end{cases}$$



$$\text{③ } f[f(x)]=\begin{cases} \ln|\ln|x||, & x \geq e^2 \\ 2\ln|x|-1, & e^2 > x \geq 1 \\ 2(2x-1)-1, & x < 1 \end{cases}$$

$$\begin{aligned} f(e^x) &= 1+x \\ f(g(x)) &= f(e^{\ln g(x)}) \\ 1+x+\ln x &= 1+\ln g(x) \end{aligned}$$

$$\begin{aligned} x+\ln x &= \ln g(x) \\ x &= \ln \frac{g(x)}{x} \end{aligned}$$

例 1.1.1

当 $x > 0$ 时 $f(e^x)=1+x$, $f(g(x))=1+x+\ln x$
 $g(x)=?$

$$\begin{aligned} \text{Solution } f(e^x) &= 1+\ln e^x \\ f(x) &= 1+\ln x \\ f(g(x)) &= 1+\ln g(x) = 1+x+\ln x \\ \ln g(x) &= x+\ln x \\ \frac{g(x)}{x} &= e^x \quad g(x) = xe^x \end{aligned}$$

② 函数的四种特性及结论

有界性 (指明区间)

单调性 ($x_1 < x_2$ 时, 恒有 $f(x_1) < f(x_2)$)

奇偶性 $F(x) = f(x) - f(-x)$ 奇

$F(x) = f(x) + f(-x)$ 偶

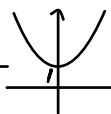
$\begin{cases} f(x) = f(2\pi-x) \\ f(\pi+x) = f(\pi-x) \end{cases}$ 关于 $x=\pi$ 对称

周期性 — $\int_a^{\pi} f(x)dx = \int_a^{\pi+T} f(x)dx$ — 奇函数 — $\int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx = 0$

数列可能用到

双曲正弦 $\frac{e^x - e^{-x}}{2}$
→ 展开也是只有奇次方

双曲余弦 $\frac{e^x + e^{-x}}{2}$
→ 展开也是只有偶次方



④ 神秘的0和1

$$f(b) - f(a) = f'(c)(b-a)$$

$$\ln(1+\frac{1}{x}) = \ln(x+1) - \ln x$$

$$f(b) - f(0) = f'(c)(b-0) \leftarrow \text{奇}$$

$$1 = e^0 \text{ 则 } e^b - e^0 = e^b - 1 \text{ 用拉氏}$$

$$\text{若 } x > 0, \text{ 证 } f(x) > f(0)x$$

$$\text{solution } \frac{f(x)}{x} > \frac{f(0)}{0} \quad f'(0) = \frac{f(x)}{x}$$

$$\text{偶函数 } f'(0) = 0$$

⑤ 祖孙三代

$F(x)$		$f(x)$		$f'(x)$
只有一个奇	$\leftarrow$	偶	可导	奇
		偶	$\leftarrow$	奇
		奇	可导	偶
T 周期	$\leftarrow$	$\int_0^T f(x) dx = 0$	可导	T 周期
有界	$\leftarrow$	有界	$\leftarrow$	有界

例 1.1.2

$$f(x) = x^2, f[\varphi(x)] = -x^2 + 2x + 3 \text{ 且 } \varphi(x) \geq 0$$

$\varphi(x)$ 及其定义域与值域.

$$f[\varphi(x)] = [\varphi(x)]^2 = -x^2 + 2x + 3 \quad \left\{ \begin{array}{l} \varphi(x) \geq 0 \\ \varphi(x) = \sqrt{-x^2 + 2x + 3} \end{array} \right.$$

$$-x^2 + 2x + 3 = -(x^2 - 2x - 3) = -(x-3)(x+1) \geq 0$$

$$x \in [-1, 3]$$

$$-x^2 + 2x + 3 \in [0, 4]$$

$$\therefore \varphi(x) \text{ 定 } [-1, 3] \text{ 值 } [0, 2]$$

$$⑥ y = \ln(x + \sqrt{x^2 + 1})$$

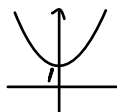
$$1. y = \ln(x + \sqrt{x^2 + 1}) \text{ 是奇函数}$$

$$2. \left. \begin{array}{l} y = \ln(x + \sqrt{x^2 + 1}) \\ -y = -\ln(x + \sqrt{x^2 + 1}) \end{array} \right\} \rightarrow \text{反函数 } x = \frac{e^y - e^{-y}}{2}$$

$$3. [\ln(x + \sqrt{x^2 + a^2})]' = \frac{1}{\sqrt{x^2 + a^2}}$$

$$4. \text{双曲正弦 } \frac{e^x - e^{-x}}{2} \rightarrow \text{展开也是只有奇次方}$$

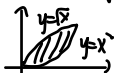
$$\text{双曲余弦 } \frac{e^x + e^{-x}}{2} \rightarrow \text{展开也是只有偶次方}$$



二. 函数的图像

二. 一 直角坐标系

- ① 常数函数
- (1) 偶函数
 - (2) 用 $y=A$ 找交点个数
 - (3) $\{y | g(x) \leq y\}$ — 概率论

- ② 幂函数
- (1)  $S = \frac{1}{3}$

(2) $\sqrt[n]{u}$ 最值: 用 u

(3) $|u|$ 最值. 用 u^2

(4) $\frac{1}{u}$ 最值 用 u

(5) 多项乘除开方乘方 取 $\ln$

- ③ 指数函数
- (1) $\lim_{x \rightarrow \infty} e^x$ 不存在 注意 $+\infty, -\infty$ 都要算

(2) $\int_{-\infty}^0 e^x = 1$ 

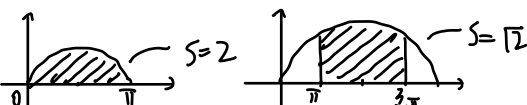
(3) $\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = 0$

$\lim_{x \rightarrow 0^+} x^\alpha \ln x = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\alpha x^{-(\alpha+1)}} = \lim_{x \rightarrow 0^+} -\frac{1}{\alpha} x^{\alpha+1} = 0 \quad (\alpha > 0)$

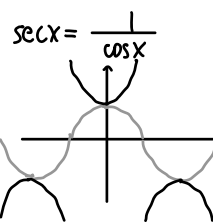
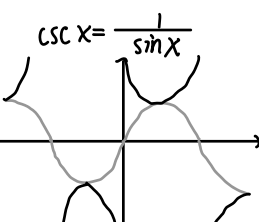
(4) $u^v = e^{v \ln u} \quad x^x = e^{x \ln x}$

$\lim u^v = \lim v(u-1)$

④ 三角函数

(1) 
 $\Rightarrow \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} |\sin x| dx = 2 - \sqrt{2}$

(2) $|\sin x| \leq 1, |\cos x| \leq 1$ 放缩时, 放成 1

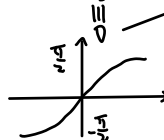
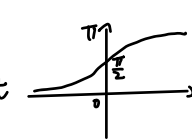
(3) $\sec x = \frac{1}{\cos x}$  $\csc x = \frac{1}{\sin x}$ 

(4) $\arcsin x + \arccos x = \frac{\pi}{2} \quad (-1 \leq x \leq 1)$

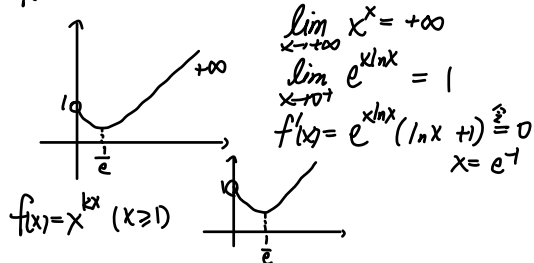
1. 证 $f(x) = \text{常数}$

$f(x) = \arcsin x + \arccos x$	$f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$
① $f'(x) = 0$	$f(0) = \frac{\pi}{2}$
② $f'(x) = A$	

2. 拉 $f(b) - f(a) = f'(p)(b-a) \equiv 0$

(1)  $\arctan$ 

1b) $f(x) = x^x$ 的大致图形 ($x > 0$)



⑤ 分段函数

$y = \text{sgn}(x)$ 符号函数 $\text{sgn} x = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

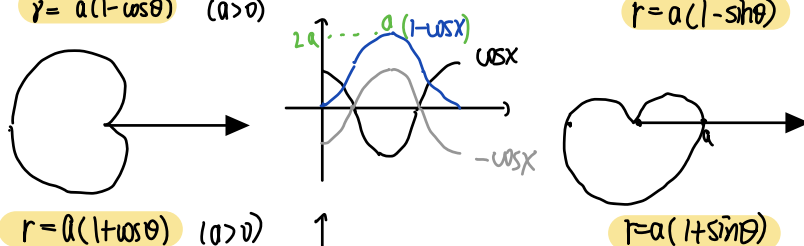
取整函数 $x-1 < [x] \leq x$

$\lim_{x \rightarrow 0^+} [x] = 0 \quad \lim_{x \rightarrow 0^-} [x] = -1$

⑥ 三种变换 平移, 对称.

二. = 极坐标系下的图像

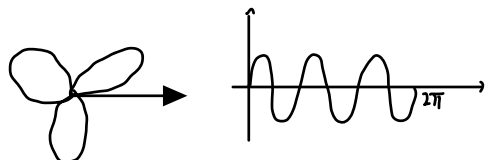
① 心形线 $r = a(1 - \cos \theta)$ ($a > 0$)



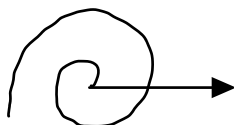
$r = a(1 + \cos \theta)$ ($a > 0$)

$r = a(1 + \sin \theta)$

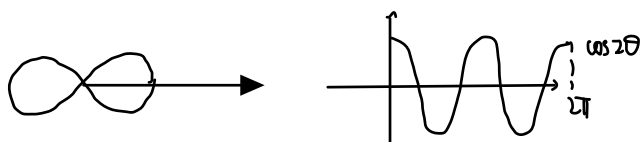
② 玫瑰线 $r = a \sin 3\theta$



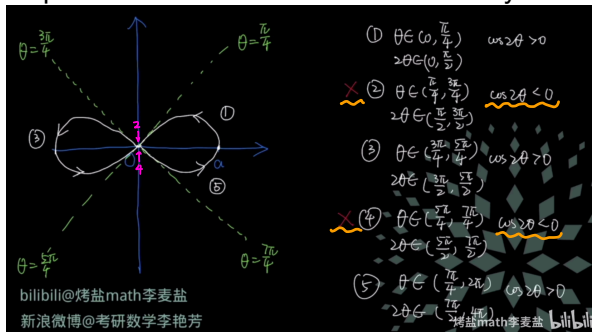
③ 阿基米德螺旋线 $r = a\theta$



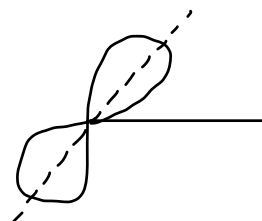
④ 伯努利双扭线 $r^2 = a^2 \cos 2\theta$ ($a > 0$)



<https://www.bilibili.com/video/BV1NC4y1W7d4>



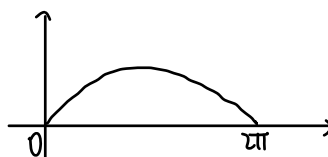
$$r^2 = a^2 \sin 2\theta$$



二. 三 参数法

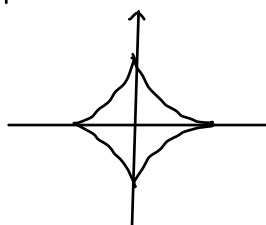
① 摆线
平摆线

$$\begin{cases} x = r(t - \sin t) \\ y = r(1 - \cos t) \end{cases}$$



② 星形线
内摆线

$$\begin{cases} x = r \cos^3 t \\ y = r \sin^3 t \end{cases}$$



三 常用基础知识

1. 数列 $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$$

证明 $\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2}$ 收敛

① $a_{n+1} - a_n = \frac{1}{(n+1)^2}$ 单增

② $\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} < \frac{1}{1} + \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 + 1 - \frac{1}{n} < 2$ 有上界

2 三角函数

诱导公式小结

$$\begin{cases} \sin(\frac{\pi}{2} \pm \alpha) = \cos \alpha \\ \sin(\pi \pm \alpha) = \mp \sin \alpha \\ \cos(\frac{\pi}{2} \pm \alpha) = \mp \sin \alpha \\ \cos(\pi \pm \alpha) = -\cos \alpha \end{cases}$$

倍角和差

$$\sin 3\alpha = -4\sin^3 \alpha + 3\sin \alpha$$

$$\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha$$

$$\tan(\frac{\pi}{4} - x) = \frac{1 - \tan x}{1 + \tan x}$$

$$\tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

$$\cot 2\alpha = \frac{\cot^2 \alpha - 1}{2\cot \alpha}$$

$$\cot(\alpha \pm \beta) = \frac{\cot \alpha \cot \beta \mp 1}{\cot \alpha \pm \cot \beta}$$

3. 指数对数运算法则

△ 多项乘除开方求导数

先求对数

$$y = \sqrt[3]{\frac{(x+1)(2x-1)^2}{(4-3x)^5}} \quad \text{求 } y'$$

$$\Delta \quad \ln \sqrt{x} = \frac{1}{2} \ln x$$

$$\ln \frac{1}{x} = -\ln x$$

$$\ln(1+x) = \ln \frac{x+1}{x} = \ln(x+1) - \ln x$$

$$\ln y = \frac{1}{3} (\ln(x+1) + 2\ln(2x-1) - 5\ln(4-3x))$$

$$y' \cdot \frac{1}{y} = \frac{1}{3} \left(\frac{1}{x+1} + \frac{4}{2x-1} + \frac{15}{4-3x} \right)$$

如当 $x > 0$ 时证明 $\frac{1}{1+x} < \ln(1+x) < \frac{1}{x}$ 连续不等式. 中值定理

$$\ln(1+\frac{1}{x}) = \ln(x+1) - \ln x = f'(\xi)(x+1-x) = \frac{1}{\xi} \quad \xi \in (x, x+1)$$

$$\frac{1}{x+1} < \frac{1}{\xi} < \frac{1}{x}$$

4 一元二次方程

韦达定理 $x_1 + x_2 = -\frac{b}{a}$ $x_1 x_2 = \frac{c}{a}$

5 因式分解

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

二项式定理: $(a+b)^n = \sum_{k=0}^n C_n^k a^{n-k} b^k = C_n^0 a^n b^0 + C_n^1 a^{n-1} b^1 + \dots + C_n^n a^0 b^n$

eg $C_5^2 = \frac{5 \times 4}{1 \times 2}$

6 阶乘与双阶乘

$$(2n)!! = 2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n) = 2^n \cdot n!$$

$$(2n-1)!! = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)$$

7 常用不等式

(1) $|a \pm b| \leq |a| + |b|$
 $|a \pm b \pm c \pm \dots| \leq |a| + |b| + |c| + \dots$

$$||a| - |b|| \leq |a - b|$$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

(2) $|ab| \leq \frac{a+b}{2} \leq \sqrt{\frac{a^2+b^2}{2}}$ eg $x_{n+1} = \sqrt{x_n(3-x_n)} \leq \frac{x_n+3-x_n}{2} = \frac{3}{2}$

$$|ab| \leq \frac{a^2+b^2}{2}$$

$$|abc| \leq \frac{a+b+c}{3} \leq \sqrt{\frac{a^2+b^2+c^2}{3}}$$

例 1.1.5

证明 $f(x) = \frac{x}{1+x^2}$ 在 $(-\infty, +\infty)$ 有界

$$|f(x)| = \left| \frac{1}{x + \frac{1}{x}} \right| \leq \frac{1}{2}$$

(3) 若 $0 < a < x < b$ $0 < c < y < d$

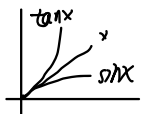
$$\text{则 } \frac{c}{b} < \frac{y}{x} < \frac{d}{a}$$

证 $n\pi < x < (n+1)\pi$ 时 $2n\pi < S(x) < 2(n+1)\pi$

$$\lim_{x \rightarrow +\infty} \frac{S(x)}{x} \rightarrow \frac{2\pi}{(n+1)\pi} < \frac{S(x)}{x} < \frac{2(n\pi)}{n\pi}$$

(4) $\sin x < x < \tan x$ ($0 < x < \frac{\pi}{2}$)

$$\sin x < x < \tan x$$

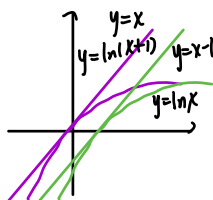


(5) $\arctan x \leq x \leq \operatorname{arcsinh} x$ ($0 \leq x \leq 1$)

(6) $e^x \geq x+1$ $x \leq e^x - 1$

(7) $x-1 \geq \ln x$

(8) $\frac{1}{1+x} < \ln(1+\frac{1}{x}) < \frac{1}{x}$ ($x > 0$)



出题

$$\sin x < x \quad \sin x_n = x_{n+1}$$

$$x < \tan x \quad x_n = \tan x_{n+1}$$

$$\arctan x \leq x \quad \arctan x_n = x_{n+1}$$

$$x \leq \operatorname{arcsinh} x \quad x_n = \operatorname{arcsinh} x_{n+1}$$

$$x \leq e^x - 1 \quad x_n = e^{x_{n+1}} - 1$$

$$\ln x \leq x - 1 \quad \ln x_n = x_{n+1} - 1$$

$$\ln(x+1) \leq x \quad \ln(x_n + 1) = x_{n+1}$$

习 1.1.1

设 $f(x) = \frac{x}{x-1}$ 验证 $f\{f[f(x)]\} = x$

并求 $f[\frac{1}{f(x)}]$, $x \neq 0, x \neq 1$

solution: $f[f(x)] = \frac{\frac{x}{x-1}}{\frac{x}{x-1} - 1} = x$

$f\{f[f(x)]\} = f\{f(x)\} = x$

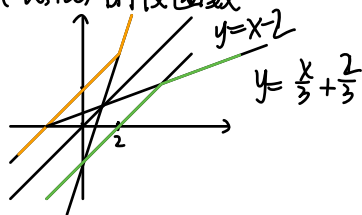
$f[\frac{1}{f(x)}] = \frac{\frac{x-1}{x}}{\frac{x-1}{x} - 1} = \frac{x-1}{-1} = 1-x, x \neq 0, x \neq 1$

习 1.2

求 $y = 2x + |2-x|, x \in (-\infty, +\infty)$ 的反函数

$y = \begin{cases} x+2 & x \leq 2 \\ 2x-2 & x > 2 \end{cases}$

$y = \begin{cases} x-2, & x < 4 \\ \frac{x}{3} + \frac{2}{3}, & x \geq 4 \end{cases}$

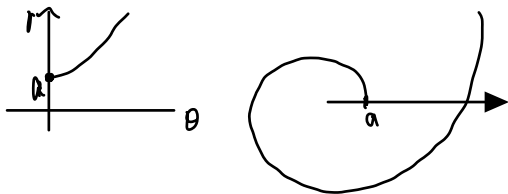


$x = \begin{cases} y-2, & y \leq 4 \\ \frac{y+2}{3}, & y > 4 \end{cases}$

$y = \begin{cases} x-2, & x \leq 4 \\ \frac{x+2}{3}, & x > 4 \end{cases}$

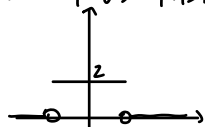
习 1.3

画出 $r = ae^{k\theta}$ ($a > 0, k > 0$) 图形



300.1.1 ①

$f(x) = \begin{cases} 2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \quad f[f(x)]$



$f(x) = \begin{cases} 2, & [-1, 1] \\ 0, & (-\infty, -1), (1, +\infty) \end{cases}$

$f[f(x)] = \begin{cases} 2, & |f(x)| \leq 1, (-\infty, -1), (1, +\infty), |x| > 1 \\ 0, & |f(x)| > 1, [-1, 1], |x| \leq 1 \end{cases}$

300.1.2

$f[\frac{x}{1+x^2}] = \frac{x+x^3}{1+x^4} \quad (\frac{x}{1+x^2})^2 = \frac{x^2}{1+x^2} + 2$
 $= \frac{\frac{x}{1+x^2} + x}{\frac{1}{1+x^2} + x^2} = \frac{(x+\frac{1}{x})}{(x+\frac{1}{x})^2 - 2} \Rightarrow f(k) = \frac{x}{x^2-2}$

300.1.3

$$f(x) = e^{x^2}, \quad f[\varphi(x)] = 1-x \quad \varphi(x) \geq 0 \quad \varphi(x)? \text{ 定义域}$$

$$f(x) = e^{x^2} \quad f[\varphi(x)] = e^{\varphi(x)^2} = 1-x$$

$$\begin{aligned} \varphi(x) &= \sqrt{\ln(1-x)} & \ln(1-x) \geq 0 & \quad x \leq 0 \\ \varphi(x) &= \sqrt{\ln(1-x)} & 1-x > 0 & \quad x < 1 \\ & \therefore x \leq 0 \end{aligned}$$

300.1.4

$$y = f(x) = \begin{cases} 1-2x^2, & x < -1 \\ x^3, & -1 \leq x \leq 2 \\ 12x-16, & 2 < x \end{cases} \quad f(x) \text{ 反函数 } g(x)?$$

$$x = \begin{cases} \frac{\sqrt{y-1}}{2}, & y < -1 \\ \sqrt[3]{y}, & -1 \leq y \leq 8 \\ \frac{y+16}{12}, & y > 8 \end{cases} \rightarrow y = \begin{cases} -\frac{\sqrt{1-x}}{2}, & x < -1 \\ \sqrt[3]{x}, & -1 \leq x \leq 8 \\ \frac{x+16}{12}, & x > 8 \end{cases}$$

300.1.5

$$f(x) = \frac{x}{\sqrt{1+x^2}}, \quad f_1(x) = f[f(x)], \quad f_2(x) = f[f_1(x)] \\ f_{n+1}(x) = f[f_n(x)] \quad n = (0, 1, 2, 3, \dots)$$

$$f_n(x) = ?$$

$$f_1 = f[f(x)] = \frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1 + \frac{x^2}{1+x^2}}} = \frac{x}{\sqrt{1+2x^2}}$$

$$f_2 = f[f_1(x)] = \frac{\frac{x}{\sqrt{1+2x^2}}}{\sqrt{1 + \frac{x^2}{1+2x^2}}} = \frac{x}{\sqrt{1+3x^2}}$$

$$f_n(x) = \frac{x}{\sqrt{1+(n+1)x^2}}$$